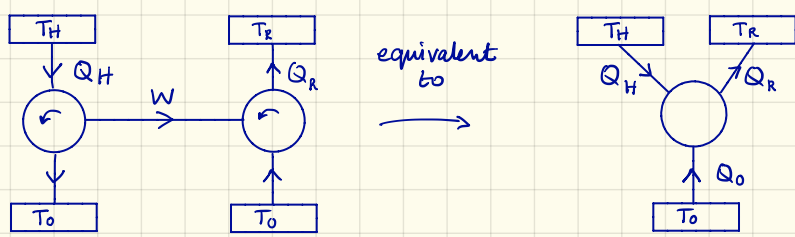


1.

(a)



1st Law : $Q_H + Q_0 = Q_R$

2nd Law : $\frac{Q_H}{T_H} + \frac{Q_0}{T_0} = \frac{Q_R}{T_R}$

Eliminate Q_0 :

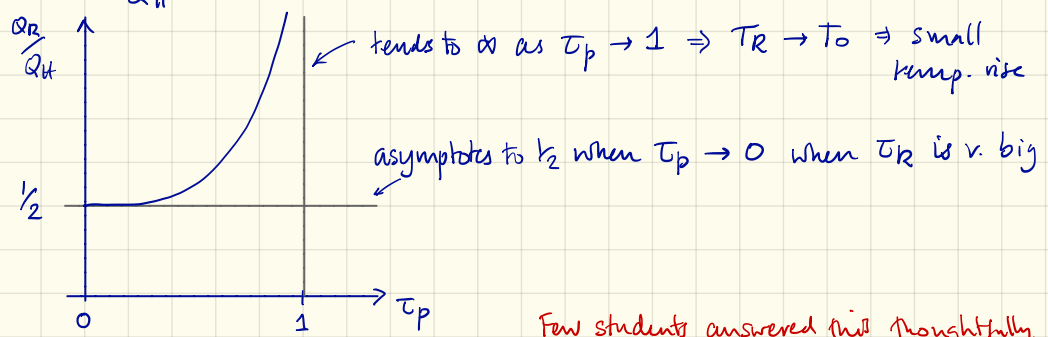
$$\frac{Q_H}{T_H} + \frac{Q_R}{T_0} - \frac{Q_H}{T_0} = \frac{Q_R}{T_R}$$

$$\Rightarrow \frac{T_0}{T_H} + \frac{Q_R}{Q_H} - 1 = \frac{T_0}{T_R} \frac{Q_R}{Q_H}$$

$$\Rightarrow \frac{Q_R}{Q_H} \left(1 - \frac{T_0}{T_R} \right) = 1 - \frac{T_0}{T_H}$$

$$\Rightarrow \frac{Q_R}{Q_H} = \frac{1 - T_E}{1 - T_P} \quad \text{where } T_E = T_0/T_H \text{ and } T_P = T_0/T_R$$

(b) $Q_R/Q_H = \frac{1}{2} (1 - T_P)^{-1}$; $T_P = T_0/T_R$ where $\infty \geq T_R \geq T_0 \Rightarrow T_P \in [0, 1]$



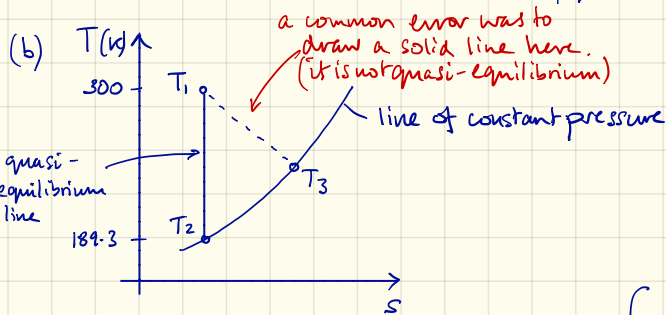
Few students answered this thoughtfully.
 The user cannot alter T_E .

(c) For max. heat input to the room, Q_R/Q_H , one wants $T_E \rightarrow 0$ (high temp. diff across engine) and $T_P \rightarrow 1$ (low temp diff across pump). T_E is fixed by power station. T_P fixed by user. $\therefore$ aim for small $T_R - T_0$.

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2.

(a) Isentropic perfect gas $\Rightarrow T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} = 300 \left(\frac{1}{10} \right)^{\frac{1}{5}} = 189.3 \text{ kelvin}$



Most students answered this question well.

(c)
$$s_3 - s_1 = c_p \ln \left(\frac{T_3}{T_1} \right) - R \ln \left(\frac{P_3}{P_1} \right)$$

$$= 1000 \left\{ \ln \left(\frac{246}{300} \right) - \frac{0.25}{1.25} \ln \left(\frac{1}{10} \right) \right\}$$

$$= 262.1 \text{ J kg}^{-1} \text{ K}^{-1}$$

$$\begin{cases} \frac{T_3}{T_1} = \frac{246}{300} ; \frac{P_3}{P_1} = \frac{1}{10} \\ R = c_p - c_v = c_p \left(\frac{\gamma-1}{\gamma} \right) \end{cases}$$

Either consider ideal gas law:

$$\frac{p_1 V_1}{T_1} = \frac{p_3 V_3}{T_3} \Rightarrow \frac{V_3}{V_1} = \frac{p_1 T_3}{p_3 T_1} = 10 \times \frac{246}{300} = 8.2$$

Or consider the gas inside the extinguisher as a system that expands against atmospheric pressure, adiabatically but not reversibly, from volume V_1 to volume V_3 . 1st law gives:

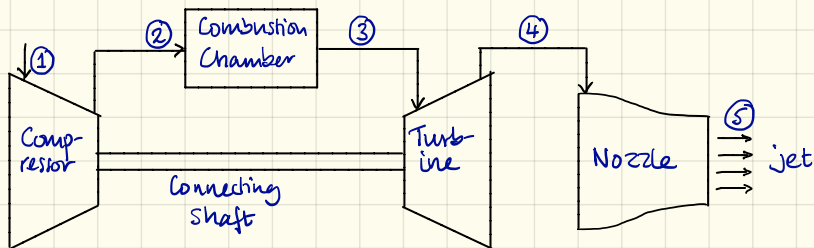
$$-p_3(V_3 - V_1) = w(T_3 - T_1)$$

$$\Rightarrow -\frac{p_3 V_1}{c_p T_1} \left(\frac{V_3}{V_1} - 1 \right) = \frac{T_3}{T_1} - 1$$

$$\Rightarrow -\frac{p_3}{P_1} \frac{R}{c_p} \left(\frac{V_3}{V_1} - 1 \right) = \frac{T_3}{T_1} - 1$$

$$\Rightarrow \frac{V_3}{V_1} = 1 - \frac{P_1}{P_3} \frac{c_p}{R} \left(\frac{T_3}{T_1} - 1 \right) = 1 - 10 \left(\frac{246}{300} - 1 \right) = 8.2$$

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$$r_p = \frac{P_2}{P_1} ; \quad \dot{Q}^* = \frac{\dot{Q}}{m c_p T_1} ; \quad \eta = \frac{\gamma-1}{\gamma}$$

(a) $T ds = dh - v dp$ ^{if p is constant} $= c_p dT$ for a perfect gas

$$\Rightarrow \int_{T_0}^T \frac{dT}{T} = \int_{s_0}^s \frac{ds}{c_p}$$

Almost all students completed
this successfully

$$\Rightarrow \ln T/T_0 = (s-s_0)/c_p$$

$$\Rightarrow T = T_0 \exp\left(\frac{s-s_0}{c_p}\right)$$

(b) S.F.E.E. in combustor : $\dot{m} c_p (T_3 - T_2) = \dot{Q} = \dot{Q}^* \dot{m} c_p T_1$
 $\Rightarrow T_3 - T_2 = \dot{Q}^* T_1$

compressor work = turbine work $\Rightarrow T_2 - T_1 = T_3 - T_4$

$$\Rightarrow T_4 = T_3 - T_2 + T_1$$

$$\Rightarrow T_4 = (\dot{Q}^* + 1) T_1$$

Most students completed
this but some tried to
use isentropic relations and
made no progress.

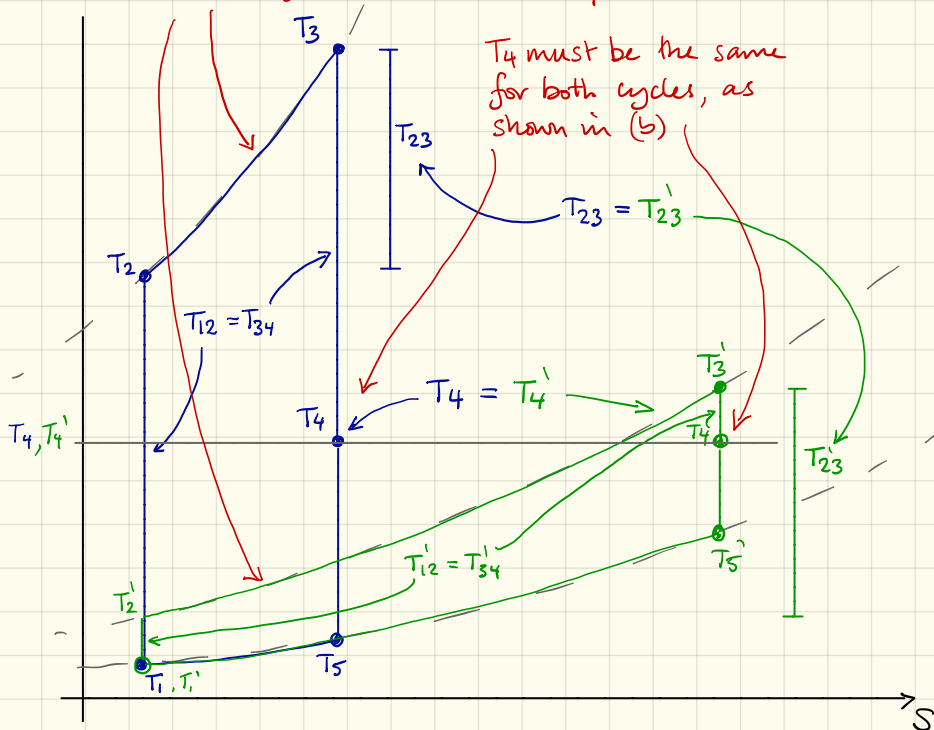
$T \uparrow$

② --- lines of const. p
(exponentials)

$T_3 - T_2$ must be the same for both cycles but $T_2 > T_2'$. This means that the low r_p cycle must extend to higher S

These lines must be exponentials, as shown in part (a)

— high r_p
— low r_p



Main points to note

- lines of constant p are exponential (part a)
- T_4 does not depend on the compression ratio (part b)
- $T_{12} = T_{34}$ for both cycles but is larger when r_p is larger
- T_{23} does not depend on the compression ratio.

There were some excellent answers but most students answered this badly. The most common error was to assume that the entropy change $S_3 - S_2$ equals $S_3' - S_2'$, even though this is incompatible with $T_4' = T_4$ and $(T_2 - T_1) = (T_3 - T_4)$ for both cycles. For 8 marks students needed to be more careful.

(d) Isentropic relation across nozzle $\Rightarrow \frac{T_4}{T_5} = \left(\frac{p_4}{p_5}\right)^\gamma$ (3)

This involves V_5^2 so must use SF.E.E.
Many students did not realise this

Assume K.E. at (4) is negligible.

SF.E.E. within nozzle $\Rightarrow \dot{m}C_p(T_4 - T_5) = \frac{1}{2}\dot{m}V_5^2$

$$M^2 = \frac{V_5^2}{\gamma R T}$$

at exit
(i.e. point 5)

$$\Rightarrow C_p(T_4 - T_5) = \frac{1}{2} M^2 \gamma R T_5$$

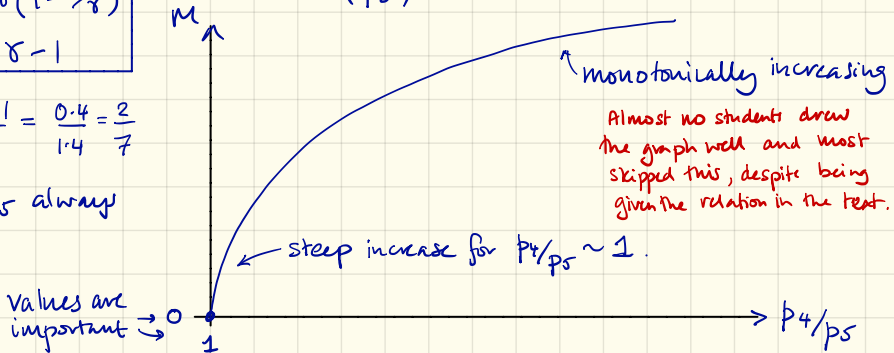
$$\Rightarrow \frac{T_4}{T_5} - 1 = \frac{1}{2} M^2 \frac{\gamma R}{C_p}$$

$$\begin{aligned} \frac{\gamma R}{C_p} &= \frac{\gamma(C_p - C_v)}{C_p} \\ &= \gamma(1 - \frac{1}{\gamma}) \\ &= \gamma - 1 \end{aligned}$$

$$\Rightarrow \left(\frac{p_4}{p_5}\right)^\gamma - 1 = \frac{1}{2} M^2 (\gamma - 1)$$

$$\eta = \frac{\gamma - 1}{\gamma} = \frac{0.4}{1.4} = \frac{2}{7}$$

$p_4 > p_5$ always



Almost no students drew the graph well and most skipped this, despite being given the relation in the text.

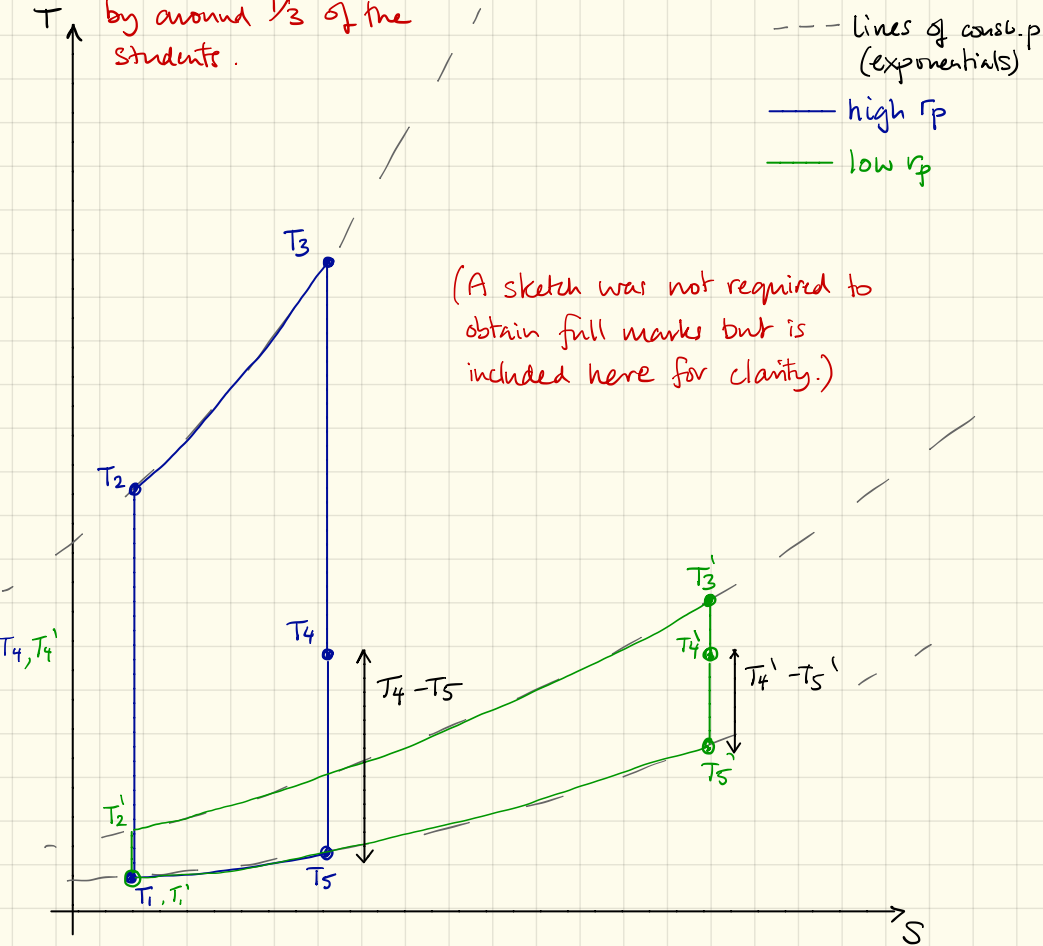
$$\begin{aligned} (e) \quad \frac{p_4}{p_1} &= \left(\frac{p_4}{p_3} \cdot \frac{p_3}{p_2} \cdot \frac{p_2}{p_1}\right) = \left(\frac{T_4}{T_3}\right)^{1/\eta} r_p \\ &= \left(\frac{1 + \dot{Q}^*}{r_p^\eta + \dot{Q}^*}\right)^{1/\eta} r_p \rightarrow r_p \text{ for } \dot{Q}^* \gg 1 \end{aligned}$$

$$\begin{aligned} T_4 &= T_1(1 + \dot{Q}^*) \\ T_3 &= T_2 + T_1 \dot{Q}^* \end{aligned} \left. \begin{array}{l} \text{from (b)} \\ \end{array} \right\} \\ &= \left(\frac{T_2}{T_1} + \dot{Q}^*\right) T_1 \\ &= (r_p^\eta + \dot{Q}^*) T_1 \end{aligned}$$

check: if $r_p = 1$, $p_4/p_1 = 1$
if $\dot{Q}^* = 0$, $p_4/p_1 = 1$

Around 10% of the students answered this well. The remainder did not.

(f) This was well answered by around $\frac{1}{3}$ of the students.



By the SFEE, $\frac{1}{2} V^2 = h_4 - h_5 = c_p(T_4 - T_5)$

$T_4 - T_5$ is greater for the high $p.$ ratio cycle, so V^2 is higher.
 $T_4 - T_5$ is greater because the entropy of the gas in the exhaust is lower, so the gas can drop to a lower temperature in the exhaust when it reaches atmospheric pressure.

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